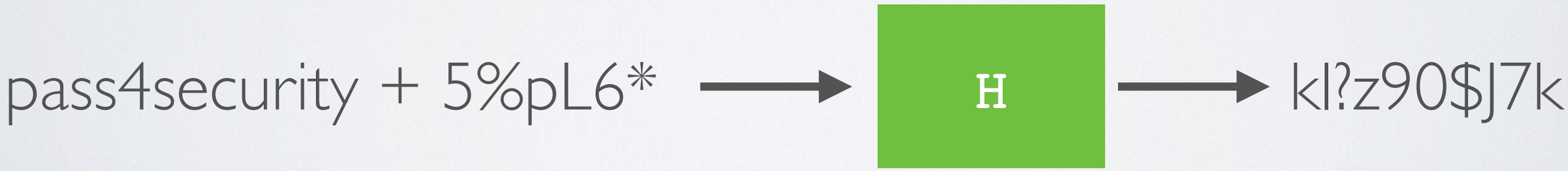
Unsalted passwords



Salted password



Specialized Password-Hashing Functions

Goal: Protect passwords against brute-force attacks

Principle: Make each guess slow i.e computationally expensive

- By including a mandatory salt
- By making the hash function CPU and memory intensive
- By having multiple passes that cannot be parallelized using GPUs or specialized hardware

◉ SHA2 & SHA3 are not suitable hash functions because too fast

✓ **Argon2** (recommended), scrypt and bcrypt